

Within the *Actinopterygii* (bony fish) lineage, marine teleosts (bony fish with mobile jaws) live in an aquatic environment where there is a higher external concentration of salt and a lower concentration of water relative to their internal concentrations. Owing to these conditions, salt ions tend to diffuse into the teleost through its skin, whereas water molecules within the organism osmotically traverse the opposite path.

Irrespective of lineage, many fish species actively assess their environment and regulate their internal concentration of fluids and electrolytes via a homeostatic process known as osmoregulation. Teleosts with either glomerular kidneys (eg, eels, sculpin) or aglomerular kidneys that are purely tubular (eg, goosefish, toadfish) have served as experimental subjects for the study of marine teleost osmoregulation. Analyzing the ionic (ie, Na^+ , Cl^- , Mg^{2+} , SO_4^{2-}) concentration of intestinal fluids, urine, and plasma in each species led to the discovery that the renal tubule of the goosefish has both excretory and reabsorptive functions.

Scientists have concluded that marine teleost osmoregulation involves the ingestion of seawater, retrieval of salts and water from the intestine, and primary excretion of divalent ions via urine and monovalent ions through the gills (thin barriers between the organism's blood and the aquatic environment).

In an experiment to further study osmoregulatory mechanisms, researchers emptied the gut of an eel and sealed the anus to prevent anal fluid loss. The eel was transported into seawater that had been treated with the volume marker phenol red, which can only be excreted anally. After 20 hours, the eel's gut contained 2.3 mL of fluid but showed a phenol red concentration equivalent to 12.3 mL of ingested seawater. Researchers found that the eel excreted 2.3 g of urine. They ultimately reported that the eel had lost all the seawater it absorbed through the gut via various mechanisms.

Scientists also study freshwater teleosts, which live in an environment where there is a higher water concentration and a lower salt concentration outside their bodies than inside. If a freshwater fish lost its ability to osmoregulate, then over time it would most likely tend to:

- ☐ A. lose water and salt. [5%]
- ☐ B. absorb water and salt. [5%]

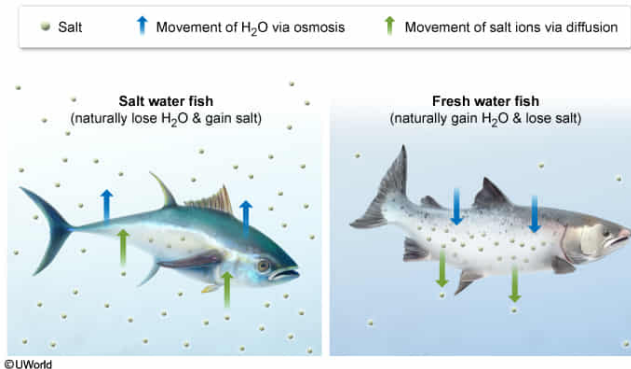
- ☐ C. absorb salt and lose water. [18%]
- ✓ ☒ D. lose salt and absorb water. [70%]

Correct

71% answered correctly

Explanation:

Diffusion of water & salt in the absence of osmoregulation



In the absence of external energy input, **molecules in solution** will **diffuse** from areas of **high concentration to low concentration**. Such flow down a concentration gradient is **thermodynamically favorable**. Molecules that are separated from an area of lower concentration by a membrane will still flow across the membrane down their concentration gradient as long as the membrane is permeable to those molecules.

The passage introduces osmoregulation, the homeostatic process by which organisms actively assess their environment and regulate their internal concentration of fluid and electrolytes. For instance, in the absence of osmoregulation, saltwater fish would tend to absorb salt and lose water, so their osmoregulatory mechanisms serve to combat this process by using energy to excrete salt and absorb water. In this way, saltwater fish can maintain the correct cellular concentrations of various molecules.

The question asks about *freshwater* teleosts. Unlike the marine teleosts described in the passage, freshwater teleosts live in an environment where there is a *higher* water concentration and *lower* salt concentration outside their bodies than inside. As a result, salt ions will tend to move to the lower-concentration environment of the exterior freshwater in the absence of osmoregulation. Water molecules will follow the opposite route and move from the higher water concentration of the external environment to the lower water concentration inside the fish (this can also be thought of as water moving from low *salt* concentration to high *salt* concentration).

(Choice A) The question states that the water concentration is greater outside of freshwater teleosts than inside, so they will *absorb* water in the absence of osmoregulation.

(Choice B) The question states that the salt concentration is lower outside of freshwater teleosts than inside, so they will *lose* salt in the absence of osmoregulation.

(Choice C) Saltwater fish, *not* freshwater fish, have a natural tendency to absorb salt and lose water.

Educational objective:

In the absence of external energy input, molecules in solution diffuse from areas of high concentration to low concentration. Osmoregulation provides the energy needed to reverse this process and maintain constant salt and water concentrations within an organism.

Time spent:0 sec

Subject:

Biology

QID:402557

System:

Digestion and Excretion

Within the *Actinopterygii* (bony fish) lineage, marine teleosts (bony fish with mobile jaws) live in an aquatic environment where there is a higher external concentration of salt and a lower concentration of water relative to their internal concentrations. Owing to these conditions, salt ions tend to diffuse into the teleost through its skin, whereas water molecules within the organism osmotically traverse the opposite path.

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In an experiment to further study osmoregulatory mechanisms, researchers emptied the gut of an eel and sealed the anus to prevent anal fluid loss. The eel was transported into seawater that had been treated with the volume marker phenol red, which can only be excreted anally. After 20 hours, the eel's gut contained 2.3 mL of fluid but showed a phenol red concentration equivalent to 12.3 mL of ingested seawater. Researchers found that the eel excreted 2.3 g of urine. They ultimately reported that the eel had lost all the seawater it absorbed through the gut via various mechanisms.

Which of the following conclusions about the eel can be made based on

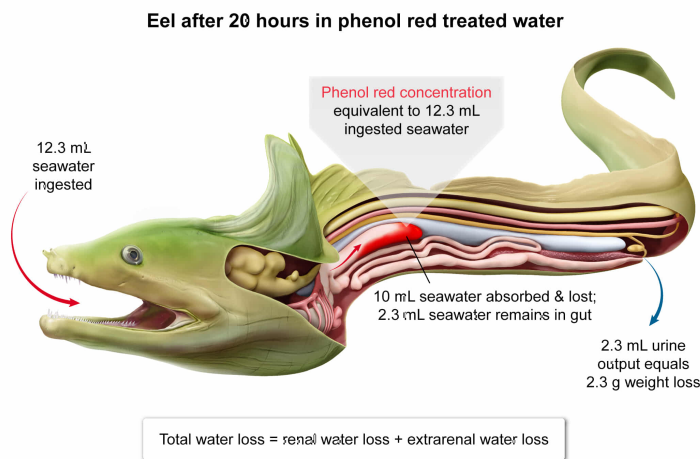
the experiment in the passage? (Note: Water density is 1.0 g/mL.)

- ☐ A. The eel absorbed 12.3 mL of water through the gut. [24%]
- ☐ B. The eel drank 10.0 mL of water. [16%]
- ✓ ☒ C. The eel lost 7.7 mL of water extrarenally. [51%]
- ☐ D. The eel excreted 4.6 mL of water renally. [7%]

Correct

52% answered correctly

Explanation:



The passage describes an experiment in which an eel's gut is emptied and it is then transferred into seawater that has been treated with phenol red, a volume marker. While the eel was in the treated water, researchers assessed the eel's gut fluid volume, gut concentration of phenol red, urine output, and weight during a 20-hour period.

In the given scenario, it is reported that after 20 hours, the eel's gut contains 2.3 mL of fluid, but the gut concentration of phenol red is equivalent to the eel having ingested 12.3 mL of water. This suggests that the eel ingested 12.3 mL of the phenol red-treated water, but then 10.0 mL were absorbed through the gut into the bloodstream ($12.3 \text{ mL} - 2.3 \text{ mL} = 10.0 \text{ mL}$). The phenol red, however, all remained in the gut because it can only be excreted via the anus. The researchers also claim that *all the water absorbed through the gut (10.0 mL) was subsequently lost*.

The passage states that the eel excreted 2.3 g of urine. Because the density of water is 1.0 g/mL, a 2.3 g weight loss is equivalent to ~2.3 mL of urine (renal water loss).

Consequently, the amount of water lost extrarenally (ie, through mechanisms that do not involve the kidneys) can be calculated using the following equation:

$$\text{Total water loss} = \text{renal water loss} + \text{extrarenal water loss}$$

$$10.0 \text{ mL} = 2.3 \text{ mL} + \text{extrarenal water loss}$$

$$10.0 \text{ mL} - 2.3 \text{ mL} = \text{extrarenal water loss}$$

$$10.0 \text{ mL} - 2.3 \text{ mL} = 7.7 \text{ mL}$$

Therefore, **the eel lost 7.7 mL of water extrarenally.**

(Choice A) The eel ingested 12.3 mL of water. However, 2.3 mL of water remained in the gut, which means that only 10.0 mL (*not* 12.3 mL)

of water was *absorbed* through the gut.

(Choice B) The eel has a gut phenol red concentration equivalent to 12.3 mL of ingested seawater; therefore, the eel drank 12.3 mL (*not* 10.0 mL) of seawater.

(Choice D) The passage states that the eel excreted 2.3 g of urine, which is ~2.3 mL, not 4.6 mL.

Educational objective:

Intestinal fluid ingestion, weight loss as water, and renal and extrarenal mechanisms of fluid excretion must be considered when calculating an organism's fluid volume.

Time spent:0 sec

Subject:

Biology

QID:402558

System:

Digestion and Excretion

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Irrespective of lineage, many fish species actively assess their environment and regulate their internal concentration of fluids and electrolytes via a homeostatic process known as osmoregulation. Teleosts with either glomerular kidneys (eg, eels, sculpin) or aglomerular kidneys that are purely tubular (eg, goosfish, toadfish) have served as experimental subjects for the study of marine teleost osmoregulation.

Analyzing the ionic (ie, Na^+ , Cl^- , Mg^{2+} , SO_4^{2-}) concentration of intestinal fluids, urine, and plasma in each species led to the discovery that the renal tubule of the goosfish has both excretory and reabsorptive functions.

Scientists have concluded that marine teleost osmoregulation involves the ingestion of seawater, retrieval of salts and water from the intestine, and primary excretion of divalent ions via urine and monovalent ions through the gills (thin barriers between the organism's blood and the aquatic environment).

In an experiment to further study osmoregulatory mechanisms, researchers emptied the gut of an eel and sealed the anus to prevent anal fluid loss. The eel was transported into seawater that had been treated with the volume marker phenol red, which can only be excreted anally. After 20 hours, the eel's gut contained 2.3 mL of fluid but showed a phenol red concentration equivalent to 12.3 mL of ingested seawater. Researchers found that the eel excreted 2.3 g of urine. They ultimately reported that the eel had lost all the seawater it absorbed through the gut via various mechanisms.

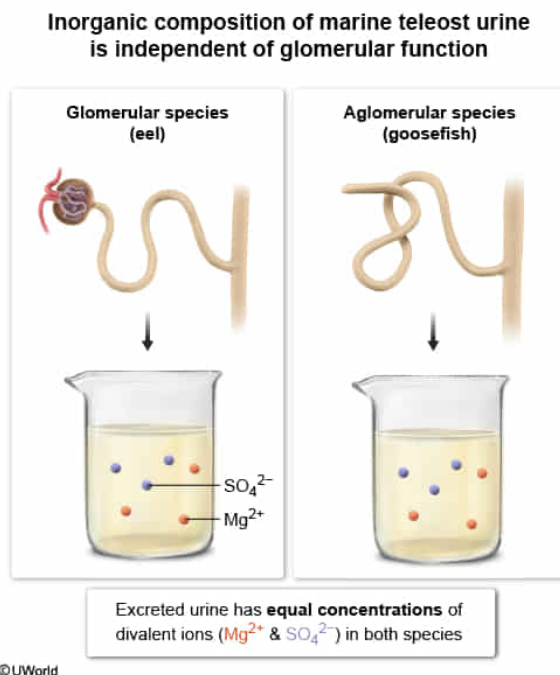
The researchers hypothesize that the ability of marine teleost kidneys to filter salts is independent of the presence of a glomerulus. Based on the passage, this hypothesis would best be supported if the researchers discovered:

- ☐ A. higher urinary concentrations of Mg^{2+} and SO_4^{2-} in the eel than in the goosfish. [6%]
- ☐ B. higher urinary concentrations of Cl^- and Na^+ in the eel than in the goosfish. [11%]
- ☒ C. equally high urinary concentrations of Mg^{2+} and SO_4^{2-} in the eel and the goosfish. [32%]
- ☐ D. equally high urinary concentrations of Cl^- and Na^+ in the eel and the goosfish. [50%]

Correct

32% answered correctly

Explanation:



According to the passage, marine teleosts such as the eel and the goosefish live in an aquatic environment that, when compared to their internal environment, has a higher salt concentration and a lower water concentration. In addition, the passage states that scientists concluded that osmoregulation in marine teleosts involves:

1. Seawater ingestion
2. Retrieval (absorption) of salt and water from the intestine
3. Excretion of **divalent ions** primarily through the **urine** and *monovalent ions* through the *gills*

In the given scenario, researchers are trying to determine whether the ability of marine teleost kidneys to filter salts depends on the presence of a glomerulus. To accomplish this, scientists studying teleosts with glomerular kidneys (eg, eels) and aglomerular kidneys (eg, goosefish) could assess how kidney structure affects urine composition in these two anatomically different fish species.

To confirm that the glomerulus does not affect filtration of salts, scientists would have to find **similar concentrations of the relevant salts** in the **urine of both glomerular and aglomerular fish**. Because **divalent ions** (eg, Mg^{2+} and SO_4^{2-}) are *primarily excreted* in the urine and monovalent ions (Na^+ and Cl^-) are *primarily excreted* through the gills of marine teleosts, researchers would have to find **equally high urinary concentrations** of the divalent ions **Mg^{2+} and SO_4^{2-}** ions in the **glomerular eel** and

the **aglomerular** goosefish.

(Choice A) Finding higher urinary concentrations of Mg^{2+} and SO_4^{2-} in the eel than in the goosefish would suggest that the glomerulus allows kidneys to filter *more* Mg^{2+} and SO_4^{2-} than kidneys that lack a glomerulus. This would indicate that salt filtration *does* depend on the presence of a glomerulus.

(Choices B and D) The passage states that monovalent ions (eg, Na^+ and Cl^-) are mainly excreted through the gills, *not* through the kidney. Therefore, measurement of monovalent ions would provide a poor assessment of kidney function.

Educational objective:

To claim that a particular function is independent of an anatomical structure, researchers must show similar function-related findings in organisms with and without the anatomical structure of interest.

Time spent:0 sec

Subject:
Biology

QID:402559

System:
Digestion and Excretion

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Analyzing the ionic (ie, Na^+ , Cl^- , Mg^{2+} , SO_4^{2-}) concentration of intestinal fluids, urine, and plasma in each species led to the discovery that the renal tubule of the goosefish has both excretory and reabsorptive functions.

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Researchers studying marine teleosts claim that the glomerular kidney is 500 times more efficient than the aglomerular kidney in eliminating nontoxic substances. Similar amounts of nontoxic ferrocyanide are intramuscularly injected into a sculpin and a goosefish of comparable weight. Given this, which of the following tables showing renal elimination of ferrocyanide after 5 hours would support the researchers' claim?

☐ A.

Species	% ferrocyanide eliminated renally
Sculpin	0.1
Goosefish	0.005

[4%]

☐ B.

Species	% ferrocyanide eliminated renally
Sculpin	10
Goosefish	0.05

[5%]

☐ C.

Species	% ferrocyanide eliminated renally
Sculpin	0.5
Goosefish	0.01

[5%]

☒ D.

Species	% ferrocyanide eliminated renally
Sculpin	5
Goosefish	0.01

[83%]

Correct

82% answered correctly

Explanation:

The **nephron** is the functional unit of the kidney, composed of a glomerulus and tubular segments. The **glomerulus**, a ball-like network of capillaries, filters blood plasma into the tubular segments of the nephron. The filtered plasma, known as the renal filtrate, enters the tubular segments, where the concentration of nutrients, electrolytes, water, and other substances is regulated via excretory and secretory processes to ultimately produce urine.

Based on the passage, some fish species have glomerular kidneys (eg, sculpin) and some have aglomerular kidneys (eg, goosefish). The **researchers claim** that the **glomerular kidney is 500 times more efficient than the aglomerular kidney** in eliminating nontoxic substances (eg, ferrocyanide). In the given scenario, a sculpin and a goosefish of comparable weight were injected with similar volumes of nontoxic ferrocyanide. The question asks for a result in the glomerular sculpin and aglomerular goosefish that would support the researchers' claim regarding kidney efficiency in glomerular and aglomerular species.

The only table that supports their hypothesis is Choice D, which shows that 5 hours after injection, the sculpin eliminated 5 percent of the ferrocyanide ion whereas the goosefish eliminated only 0.01 percent. The exact level of increased ferrocyanide elimination in the glomerular sculpin compared to the aglomerular goosefish can be calculated as follows:

$$\frac{\% \text{ ferrocyanide eliminated by glomerular sculpin}}{\% \text{ ferrocyanide eliminated by aglomerular goosefish}} = \frac{5\%}{0.01\%} = \frac{\frac{5}{1}\%}{\frac{1}{100}\%}$$

Dividing one fraction by another fraction is equivalent to multiplying the first fraction by the *reciprocal* of the second fraction, so the equation can be rewritten as:

$$\frac{\frac{5}{1}}{\frac{1}{100}} = \frac{5}{1} \times \frac{100}{1} = 500$$

Therefore, the results in this table show that the glomerular kidney is **500 times more efficient** in **eliminating ferrocyanide** than the **aglomerular kidney**.

(Choice A) Calculating $0.1/0.005$ would show that the glomerular kidney is 20 times more efficient in eliminating ferrocyanide as the aglomerular kidney.

(Choice B) Calculating $10/0.05$ would show that the glomerular kidney is 200 times more efficient in eliminating ferrocyanide as the aglomerular kidney.

(Choice C) Calculating $0.5/0.01$ would show that the glomerular kidney is 50 times more efficient in eliminating ferrocyanide as the aglomerular kidney.

Educational objective:

As the functional unit of the kidney, the nephron consists of a ball-like network of capillaries called the glomerulus. The glomerulus filters blood plasma into the tubular segments of the nephron.

Time spent: 0 sec

Subject:

Biology

QID: 402560

System:

Digestion and Excretion